

ALIGUDARZ, Iran, Islamic Republic of

WMO: 407830

Lat: 33.4080N

Lon: 49.7032E

Elev: 2022

StdP: 79.28

Time Zone: 3.50 (IRN)

Period: 96-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-12.1	-9.2	-19.4	0.9	-3.2	-17.0	1.1	0.1	14.8	2.4	12.4	3.8	5.5	120	0.593

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.9	34.5	14.7	33.5	14.3	32.5	13.9	15.7	32.2	15.1	31.6	14.5	30.8	5.1	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
9.2	9.2	19.3	8.0	8.5	18.7	7.1	8.0	17.8	51.3	32.3	49.3	31.5	47.5	31.1	20.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.1	10.3	9.1	DB	-15.1	37.6	3.7	1.7	-17.7	38.8	-19.9	39.8	-21.9	40.8	-24.6	42.0	(4)
(5)			WB	-15.3	17.2	3.9	1.3	-18.2	18.1	-20.5	18.8	-22.7	19.5	-25.6	20.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	13.0	-0.6	2.3	7.1	11.0	16.5	22.3	25.7	25.2	20.8	14.6	7.2	2.8		
	DBStd	9.46	4.78	4.10	3.77	3.29	2.61	2.37	2.06	1.82	2.15	2.88	3.37	3.97		
	HDD10.0	994	328	217	104	26	0	0	0	0	0	2	95	223		
	HDD18.3	2611	586	449	347	220	68	2	0	0	4	119	334	481		
	CDD10.0	2083	0	1	15	57	201	371	488	471	324	144	11	1		
	CDD18.3	657	0	0	0	0	11	122	230	212	79	3	0	0		
	CDH23.3	8321	0	0	0	5	153	1567	2904	2635	1009	46	0	0		
	CDH26.7	3652	0	0	0	0	12	626	1458	1260	296	1	0	0		
(14) Wind	WSAvg	4.5	4.3	4.4	4.6	4.4	4.5	4.5	4.8	4.9	4.4	4.4	4.0	4.3		
(15) Precipitation	PrecAvg	432	56	53	69	75	28	3	2	1	2	24	45	71		
	PrecMax	609	117	158	151	179	120	16	31	5	23	91	205	158		
	PrecMin	276	18	12	4	2	1	0	0	0	0	0	0	17		
	PrecStd	92	24	33	40	45	26	4	6	1	5	27	39	41		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.1	14.4	20.7	24.0	28.2	34.4	36.2	35.4	31.8	26.0	19.2	15.1		
		MCWB	3.3	5.0	7.6	9.3	12.3	14.9	15.5	14.8	12.9	11.1	7.6	5.2		
	2%	DB	9.4	11.9	18.2	21.0	26.5	33.0	34.9	34.2	30.7	24.7	17.0	12.6		
		MCWB	2.8	3.5	6.4	9.5	11.7	14.1	14.9	14.3	12.7	10.3	6.7	4.3		
	5%	DB	7.7	10.3	16.0	19.2	25.1	31.6	33.9	33.2	29.7	23.3	15.3	10.7		
		MCWB	2.0	3.0	5.6	8.6	11.2	13.5	14.6	14.1	12.4	9.7	6.2	3.6		
	10%	DB	6.2	8.5	14.1	17.7	23.7	30.2	32.9	32.2	28.5	21.9	13.4	8.9		
		MCWB	1.6	2.5	4.8	7.9	10.9	12.9	14.2	13.6	11.9	9.1	5.5	3.0		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	5.0	6.2	8.5	11.2	14.0	16.2	16.7	16.0	14.4	12.0	9.2	6.8		
		MCDB	8.5	11.0	16.9	20.3	25.1	30.2	33.4	32.7	29.7	23.2	15.1	11.6		
	2%	WB	3.9	5.0	7.2	10.2	12.9	15.1	15.9	15.4	13.5	11.1	8.1	5.5		
		MCDB	7.4	10.2	14.6	18.6	24.0	30.6	32.8	32.0	28.7	22.5	14.4	10.4		
	5%	WB	3.0	3.9	6.4	9.4	12.2	14.3	15.2	14.9	12.9	10.4	7.2	4.5		
		MCDB	6.3	8.6	13.7	17.4	22.7	29.7	32.0	31.4	28.2	21.3	13.0	9.2		
	10%	WB	2.0	2.9	5.7	8.7	11.5	13.6	14.7	14.3	12.3	9.7	6.3	3.5		
		MCDB	5.8	7.6	12.7	16.0	21.6	28.9	31.3	30.8	27.3	19.9	11.9	7.8		
(35) Mean Daily Temperature Range	5% DB	MDBR	8.9	9.3	10.3	10.6	12.2	14.9	13.9	14.2	15.2	12.7	10.1	9.4		
		MCDBR	10.3	11.6	12.7	12.5	14.4	16.3	15.1	15.3	16.0	14.5	12.8	11.8		
	5% WB	MCWBR	6.1	6.2	5.7	5.2	5.7	6.6	6.0	6.2	7.1	7.0	6.3	6.5		
		MCWBR	8.3	9.6	10.2	11.1	13.3	15.3	14.0	14.0	15.2	13.1	10.0	9.9		
(40) Clear-Sky Solar Irradiance	taub		0.274	0.317	0.378	0.472	0.461	0.435	0.482	0.438	0.368	0.379	0.319	0.275		
	taud		2.426	2.297	2.097	1.872	1.925	2.001	1.892	2.003	2.190	2.132	2.334	2.464		
	Ebn at Noon		950	932	892	821	834	852	810	842	891	845	880	929		
	Edh at Noon		90	115	152	200	192	177	197	173	137	134	99	82		
(44) All-Sky Solar Radiation	RadAvg		2.99	3.88	5.08	5.84	7.00	8.30	7.75	7.31	6.44	4.74	3.30	2.69		
	RadStd		0.23	0.37	0.37	0.49	0.50	0.31	0.35	0.29	0.24	0.32	0.30	0.31		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days					
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(47) Regional (7 neighbors)		N/A	N/A	N/A	+0.72	-0.85	-1.21	N/A	N/A	N/A	N/A		

Nomenclature: See separate page